Sarbajit Ghosh

CONTACT INFORMATION

Department of CSE
IIT Kanpur

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Publications: Google Scholar

CAREER VISION

My research interest aligns with the interplay between **differential privacy** and **machine learning**. I like to apply my interdisciplinary background, as well as my abilities, to find creative solutions to complex problems. This way, I like to deeply engage myself in advancing the subject to make a real-world impact. My other interests lie in the broader spectrum of **privacy-preserving technologies** and **statistics**.

ACADEMIC BACKGROUND

Ph.D (Pursuing) in Computer Science 2021 - Present
Indian Institute of Technology, Kanpur
CPI: 8.00

M.Tech in Cryptology and information security 2019 - 2021
Indian Statistical Institute, Kolkata
Marks: 81.35%
First Division with Distinction

GATE Qualified In Statistics 2019
Percentile: 98.01

GATE Qualified In Computer Science 2021
Percentile: 87.26

M.Sc (Post Graduation) in Statistics 2016 - 2018
West Bengal State University, Barasat
Marks: 81.3%
Recipient of First-Class First Vice-Chancellor Gold Medal

B.Sc(Graduation) in Mathematics 2012 - 2016
Rishi Bankim Chandra College, under West Bengal State University
Marks: 54.63%

Higher Secondary Examination 2009 - 2011
Naihati Narendra Vidyaniketan, under WBCHSE
Marks: 72.6%

Secondary Examination 2009
Ramkrishna Vivekananda Misson, under WBBSE
Marks: 79.38%

PUBLICATIONS

1. A Generalized Unknown Malware Classification. Rani, N., Mishra, A., Kumar, R., **Ghosh, S.**, Shukla, S.K. and Bagade, P. International Conference on Security and Privacy in Communication Systems (SecureComm), 2022

TEACHING ACTIVITIES

1. **Teaching Assistant, CSE, IIT Kanpur**
   - CS729 & CS798 Differential Privacy in Machine Learning — Spring 2026, 2025
   - CS787 Generative Artificial Intelligence — Fall 2025

- CS685 Data Mining — Fall 2024, 2021
- CS689 Computational Linguistics for Indian Languages — Spring 2024, Fall 2022
- CS798 Introduction to Probability for Computer Science — Fall 2023
- ESC112 Fundamentals of Computing - II — Summer 2023
- CS315 Principles of Database Systems — Spring 2023
- CS657 Information Retrieval — Spring 2022

2. **Guest Lecturer at the Department of Statistics, Chakdaha Day College, Kalyani University** August 2018 – July 2019
   - **Responsibilities:** Teaching, Organising internal & external examinations
   - **Courses taught**: Descriptive statistics & probability theory, Statistical inference at UG level

Internship

1. **Statistical Programmer at Imerit Technologies, Kolkata, India** June 2020 - August 2020
   - **Project Name:** An Analysis of Factors that Impact Data Annotation Productivity in Complex CV Projects
   - **Project Domain: Statistical Analysis**, **Socio-economic Survey**
   - **Description of Project:** To model the time per task, in CV projects of an employee, based on project characteristics and some external parameters such as training given by the company, socio-economic profile of that employee, etc as a part of a team
   - **Technologies Used: R** packages **dplyr**, **ggpolt2**, **caret**, **Google sheets**, **Tableau**, NoSQL database **MongoDB**

Academic Projects

1. **Internet voting using Blockchain** Spring 2021
   Guide: Dr. Souradyuti Paul, IIT Bhilai
   - **Project Domain: Secure Multiparty computation**, **Zero Knowledge Proofs**, **Blockchain**, **Internet voting**
   - **Description of Project:** Application of Ethereum blockchain, in designing a complete decentralized voting protocol together with related security proofs
   - **Technologies Used: Ganache**, **Truffle**,**MetaMask**, **Solidity**, **W3.JS**

2. **A short survey on Differential Privacy** Fall 2020
   Supervisor: Dr. Srimanta Bhattacharya, R. C. Bose Centre for Cryptology and Security

3. **Secure Banking Website** Spring 2020
   Supervisor: Dr. Nilanjan Datta, ISI Kolkata
   Project Purpose: To create a banking website which is secure against **SQL injection**, **CSRF** and **Clickjack attack**

4. **A note on How to Access Program Counter and Instruction Register by C programming** Fall 2019
   Supervisor: Prof. Subhamoy Maitra, ISI Kolkata

5. **Robust model-based adjustments to GREG estimation when some responses are missing** Spring 2018
   Supervisor: Dr. Sumanta Adhya, WBSU
   Project Purpose: To study the robustness property of the GREG estimator in different cases when imputation was done by parametric and non-parametric regression and when data size is small

Programming Skill

- **Languages:** C, Python, R, HTML
- **Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, Opacus, NLTK, spaCy
- **Database & Publishing:** MySQL, LaTeX, Markdown
- **OS:** Linux

Personal Details

- **Languages:** Bengali (Mother tongue), English, Hindi
- **Gender:** Male
- **Nationality:** Indian
- **Hobbies:** Violin, Indian Classical Music, Regional History